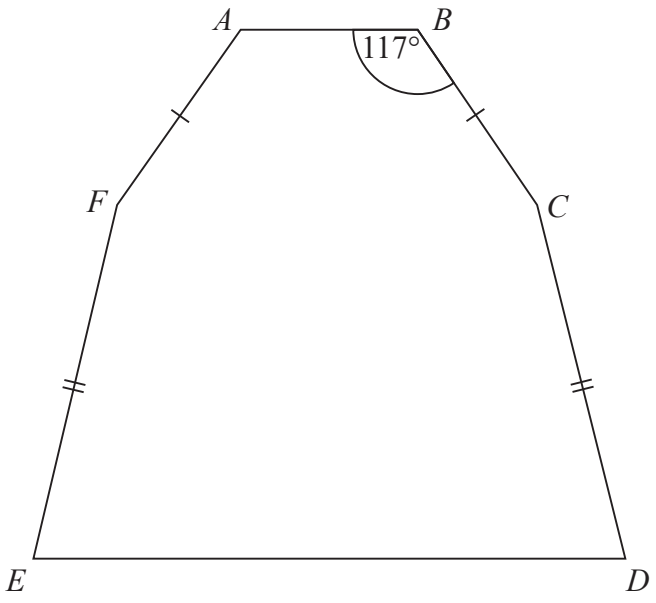
5 The diagram shows a hexagon.
The hexagon has one line of symmetry.



$FA = BC$
$EF = CD$
Angle $ABC = 117°$

Angle $BCD = 2 \times$ angle CDE

Work out the size of angle AFE.
You must show all your working.

Interior sum = (6 – 2) × 180° = 720°

1 mark
Interior angle sum 720°.

Four unknown angles add to 6x.

1 mark
Use x, x, 2x, 2x.

6x = 720° – 2 × 117° = 486°

1 mark
Solve x = 81°.

x = 81°; angle AFE = 2x = 162°

1 mark
Conclude AFE = 162°.

162 °

(Total for Question 5 is 4 marks)